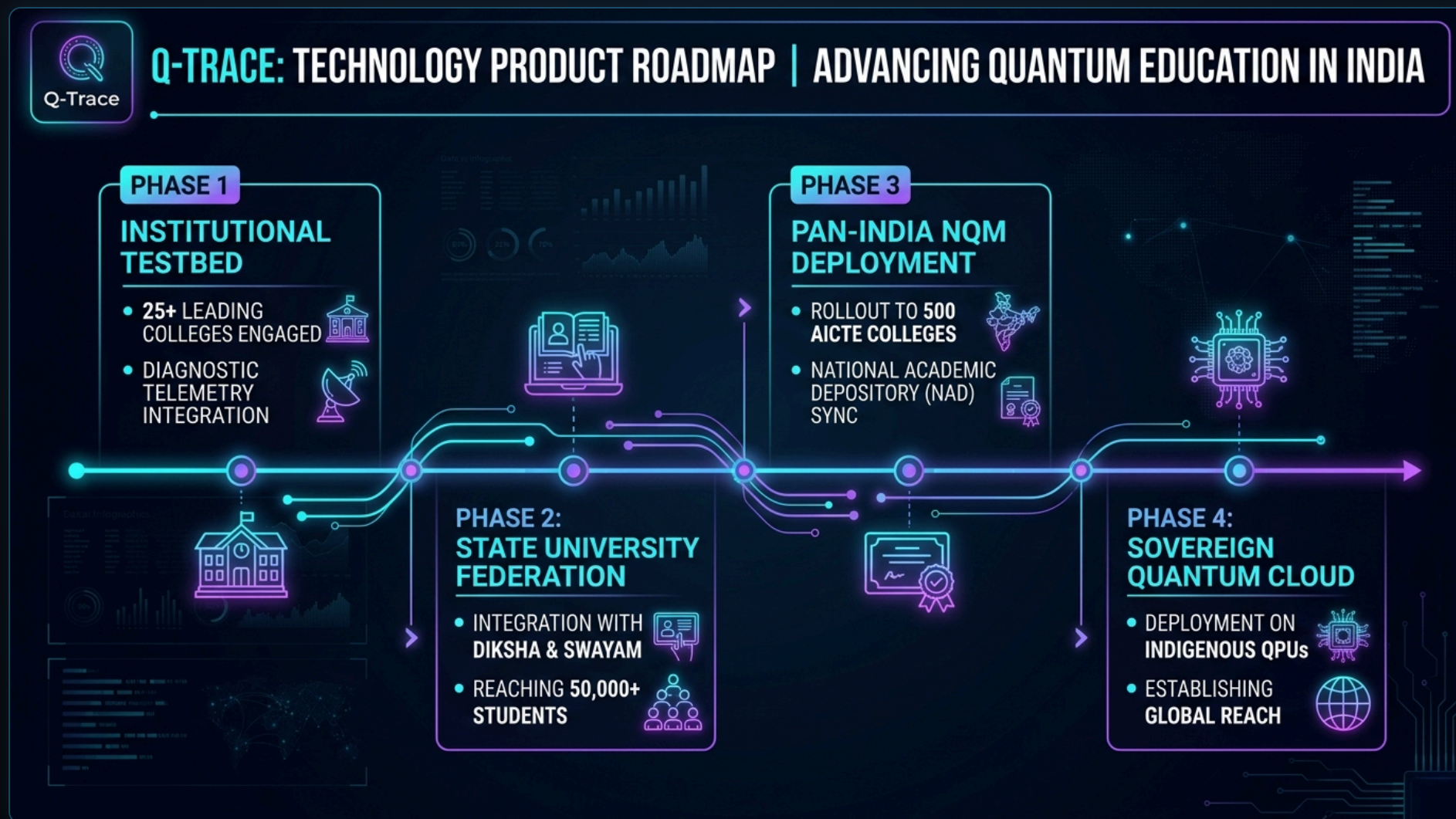




HORIZON 1

M 1–6

Institutional Testbed: 25 AICTE Pilot Colleges, Diagnostic Telemetry Calibrated, AST Security Sandbox.

HORIZON 2

M 6–18

State Federation: AKTU/VTU/Anna Univ, SWAYAM & DIKSHA Integration, 50,000 Active Learners.

HORIZON 3

M 18–30

Pan-India NQM: 500+ Institutions on NIC Cloud, DigiLocker/NAD Sync, Multi-Lingual Socratic AI.

HORIZON 4

M 30–48

Sovereign Cloud: Indigenous Indian QPU Adapters, 25,000 Certified Quantum Talent Target.



DETAILED PHASE SPECIFICATIONS & DELIVERABLE MATRIX

Technical Deliverables, Regulatory Accreditations, and Verified National Economic Impact

VERIFIED IMPLEMENTATION PLAN

NATIONAL SCALE

Atmanirbhar Bharat • Viksit Bharat 2047 Alignment

PHASE 1: Institutional Testbed

Months 1–6

Objective: Deploy across 25 select AICTE technical colleges engaging 5,000 active learners.

- **Technical:** Quantum Flight Recorder, closed AST sandbox, 40+ Misconception Decision Rules.
- **Faculty:** Instructor Failure Topology Dashboard for automated 1-click cohort remediation.
- **Compliance:** GeM (Govt e-Marketplace) catalog listing, AICTE NEAT 4.0, CERT-In audit.

Target: 60% relative boost in quantum conceptual reasoning

PHASE 2: State University Federation

Months 6–18

Objective: Federation with State Technical Universities (AKTU, VTU, Anna Univ) scaling to 50,000 learners.

- **Integration:** SWAYAM/NPTEL and DIKSHA syndication; NKN edge caching for <10ms latency.
- **Curriculum:** Tri-Engine Arena (Qiskit Aer, PennyLane, Cirq); Shor, QAOA, and VQE templates.
- **Finance & Ops:** Operating cash-flow breakeven; AICTE ATAL FDPs certifying 1,000+ faculty.

Target: ₹12+ Crore saved in physical college lab setups

PHASE 3: Pan-India NQM Deployment

Months 18–30

Objective: Pan-India rollout across 500+ AICTE colleges supporting 100,000+ concurrent students.

- **Infrastructure:** Sovereign hosting on National Informatics Centre (NIC) MeghRaj GovCloud.
- **Credentialing:** National Academic Depository (NAD) & DigiLocker verifiable quantum badges.
- **Pedagogy:** Multi-lingual Socratic AI Tutor localized in Hindi, Telugu, Tamil, Marathi, Kannada.

Target: 4× faster algorithm mastery; 65% faculty grading reduction

PHASE 4: Sovereign Quantum Cloud

Months 30–48

Objective: Connect learner circuits directly to indigenous Indian QPUs built under NQM.

- **Hardware:** Agnostic adapters for Indian superconducting & trapped-ion QPUs (TIFR, RRI, IISc).
- **Advanced Labs:** Quantum Error Correction (QEC) interactive labs (Surface codes, Stabilizers).
- **Global Reach:** OpenQASM 3.0 compilation and international quantum research export gateway.

Target: 25,000 certified quantum engineers; ₹1,250+ Cr economic value

NATIONAL SCALE & MACRO ECONOMIC IMPACT LEDGER

NATIONAL SCALE METRIC	YEAR 1 (PILOT PHASE)	YEAR 2 (FEDERATION)	YEAR 3 (NQM SCALE)	YEAR 4 (QPU HORIZON)
Participating Technical Colleges	25 Colleges	150 Colleges	500 Colleges	1,200+ Colleges
Active Trained Learners	5,000 Students	50,000 Students	100,000+ Students	250,000+ Students
Institutional Lab Cost Saved	₹3.75 Crore	₹22.50 Crore	₹75.00 Crore	₹180.00+ Crore
Certified Quantum Graduates (NQM Target)	1,200 Ready	8,500 Ready	18,000 Ready	25,000+ (100% Target Met)